

90. Which of the following statements is the most accurate description for the concept of Standard deviation?
- A. The total distance from the smallest score to the highest score.
  - ☒ B. The square root of the total distance from the smallest score to the highest score.
  - C. The squared average distance between all scores and the mean.
  - D. The average distance between a score and the mean.
91. A coin is tossed 5 times, what is the probability of getting exactly 3 heads?
- A. 0.5
  - ☒ B. 0.3125
  - C. 0.1563
  - D. 0.03125
92. A university found that 20% of its students withdraw without completing the introductory statistics course. Assume that 20 students registered for the course. What is the probability that two or fewer will withdraw?
- 20%  $0.02 + 20$   $0.2$
- A. 0.1369
  - B. 0.2060
  - C. 0.2054
  - ☒ D. 0.7940
  - E. None of the above
93. A manufacturer of screws has found that on an average 0.04 of the screws produced are defective. A random sample of 400 screws is examined for the proportion of defective screws. What is the probability that the proportion of defective screws in the sample is greater than 0.05?
- $a = 0.10$   $s = 400$   $0.04$
- ☒ A. 0.1539
  - B. 0.3461
  - C. 0.4793
  - D. 0.0207
  - E. None of the above.
94. If we say that  $\alpha = 0.10$  for a particular hypothesis test, we are saying that:
- A. Ten percent is the probability for accepting a hypothesis that is true.
  - B. Ten percent is the risk we take of rejecting a hypothesis that is true.
  - C. Ten percent is the risk we take of accepting a hypothesis that is false.
  - D. A and B only.
  - E. A and C only.
95. A chi-square value can never be negative because:
- ☒ A. Differences between expected and observed frequencies are squared.
  - B. A negative value would mean that the observed frequencies were negative.
  - C. The absolute value of the differences is computed.
  - D. Only A and B.
  - E. None of the above.
96. Regression and correlation methods are related to which of the following issues?
- A. Measuring the rate of change in one variable for a given change in another variable.  $r$   $r^2$
  - B. Bringing out the nature of relationship between any two variables.  $r$
  - C. Evaluating the strength of the relationship.  $r$   $r^2$
  - ☒ D. All of the above.
  - E. None of the above